What is Software Testing?

The process of **verifying and validating a software application to ensure it works as expected and is free from defects (bugs).**
Checking whether the application behaves according to the requirements.

Example:

Imagine a **login page**.

Requirement:

- User should enter **username**
- User should enter **password**
- Clicking **Login** should open the **Dashboard**

Tester checks:

1. Enter valid username and password → Dashboard opens (pass)
2. Enter wrong password → Error message should appear (pass)

If something does not work as expected → **Bug/Defect**

Why do software testing?

- Find **bugs or defects**
- Ensure **quality of the application**
- Verify the software meets **requirements**
- Ensure the system is **reliable and stable**

Manual Testing vs Automation Testing

Manual Testing is the process of testing the software **manually without using automation tools**.

All steps are done **by a human tester**.

Advantages
- Easy to start
- No programming knowledge required

Disadvantages
- Time consuming
- Repetitive work
- Higher chance of human error

Automation Testing is the process of testing software using **automation tools and scripts**.

The tester writes **test scripts** that automatically execute test cases.

Tools example: Selenium, Playwright, Robot Framework

Example:

Instead of manually logging in, a script does it automatically.

Advantages

- Faster execution
- Reusable scripts
- Ideal for **regression testing**
- Works in **CI/CD pipelines**

Disadvantages

- Requires programming knowledge
- Initial setup time
- Not suitable for testing some captcha

What is Selenium?

Selenium is an **open-source automation tool used to automate web browsers for testing web applications.**

Components of Selenium

1. **Selenium IDE:**

Selenium IDE is a **record-and-playback tool** used to create automation tests without programming.

2. **Selenium WebDriver**

Selenium WebDriver is the **most important component of Selenium**.
It allows testers to **write scripts using programming languages to control browsers.**

3. **Selenium Grid**

Selenium Grid is used to **run tests on multiple browsers and machines simultaneously.**
Example:
Run the same test on: Chrome, Firefox, Edge
at the **same time**.

4. **Selenium RC (Remote Control)**

Selenium RC is the **old version of Selenium**.
It allowed automation before WebDriver existed.
Now it is **deprecated and no longer used**.
WebDriver replaced Selenium RC.

Selenium Architecture Flow

Client Library → W3C Protocol → WebDriver → Browser

Communication Flow

- Client Library sends **HTTP Request**
- W3C Protocol transfers the request
- WebDriver communicates with the browser
- Browser sends **Response back through WebDriver**
- WebDriver sends **HTTP Response to Client Library**

Client Library is the **platform where we write automation scripts**.
W3C stands for **World Wide Web Consortium**.
It is used to **transfer data between the client and the server**.
Data is transferred through **HTTP requests and responses**.
It acts like a **translator between the Client Library and the Browser Driver**.